



Dwight Look College of

ENGINEERING

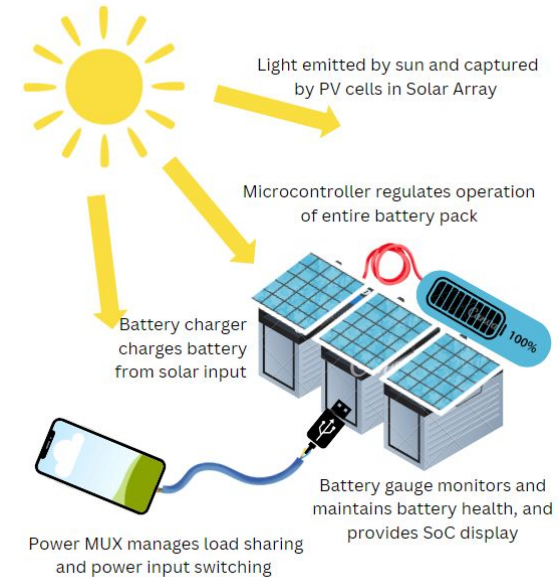
TEXAS A&M UNIVERSITY

Team 64: TI Solar Battery Pack Bi-Weekly Update 2

**Jaein Cha, Patrick Wang, Cole Miller, Camden Pomeroy
Sponsor: Wyatt Keller**

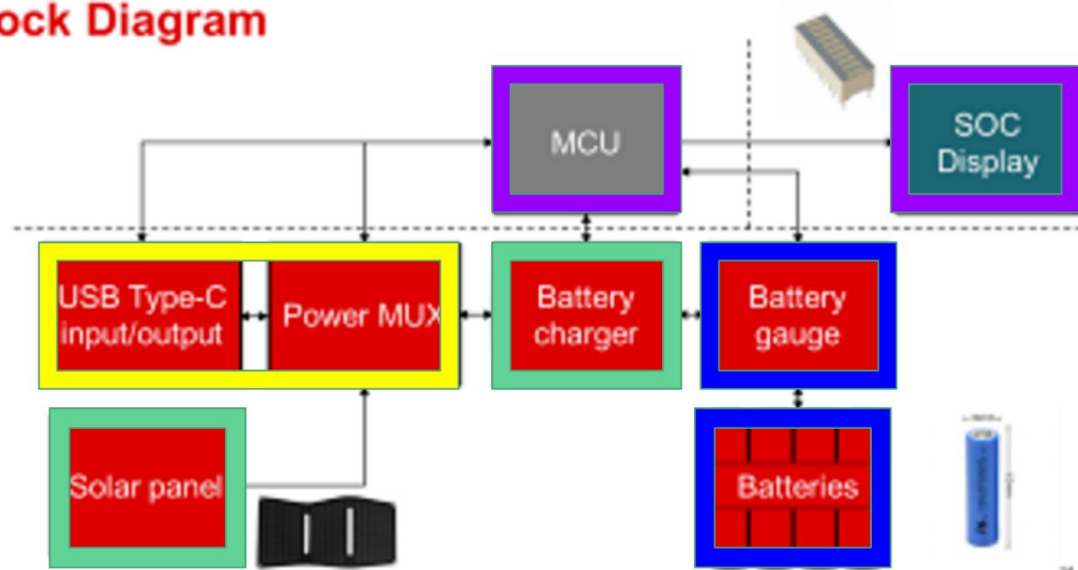
Project Summary

- Natural disasters can cause unexpected power outages resulting in the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.



Project/Subsystem Overview

Block Diagram



Camden ■
 Cole ■
 Patrick ■
 Jaein ■



Project Timeline



January

Individual Subsystem Finalization

Wrap up subsystem design and
validation testing



February

Integration Design

Work together to integrate the
designs of each subsystem



March

Assembly/Validation

Assemble the final pcb and 3D
printed enclosure. Perform
validation testing and identify
problems that occur



April

Validation/Debugging

Engineer a solution to problems
that arise as a result of
integration



Battery Gauge / Battery Pack

Patrick Wang

Accomplishments since last presentation 21 hrs	Ongoing progress/problems and plans until the next presentation
• Soldered new board, I2C communication works between board and bqStudio • Programmed IC calibration to read specific parameters for the INR18650 30Q batteries we are using	• Currently trying to turn on FETs to enable power path for charging • Start integrating with charger subsystem

Battery Gauge / Battery Pack

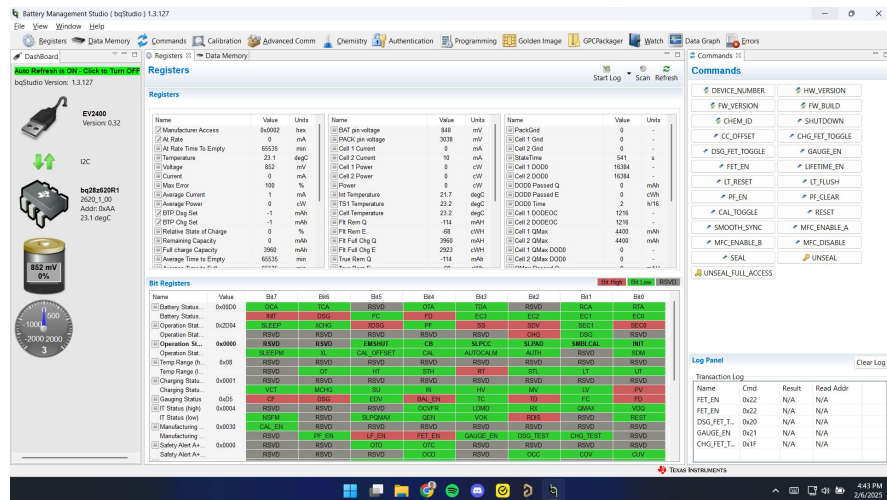
Patrick Wang

What works:

- Board connects to bqStudio application through successful I2C connection
- Can successfully read battery parameters while connected to charger load

What doesn't work:

- Need to open FETs to allow charging in 2.5-4.3V range



Bit Registers										Bit High	Bit Low	RS
Name	Value	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0			
Battery Status...	0x00D0	OCA	TCA	RSVD	OTA	TD	EC3	EC2	EC1	RTA		
Battery Status...	0x00D0	INIT	DSG	FC	FD	EC3	EC2	EC1	EC0			
Operation Stat...	0x2D04	SLEEP	XCHG	XDSG	PF	SS	SDV	SEC1	SEC0			
Operation Stat...	0x0000	RSVD	RSVD	RSVD	RSVD	RSVD	CHG	DSG	RSVD			
Operation Stat...	0x0000	RSVD	RSVD	EMSHUT	CB	SLPCC	SLPAD	SMBLCL	INIT			
Operation Stat...	0x0000	SLEEP	XL	CAL_OFFSET	CAL	AUTOCALM	AUTH	RSVD	SDM			
Temp Range (h...	0x08	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD			
Temp Range (l...	0x08	RSVD	OT	HT	STH	RT	STL	LT	UT			
Charging Status...	0x0001	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD			
Charging Status...	0x0001	VCT	MCHG	SU	IN	HV	MV	LV	PV			
IT Status (low)	0x0004	CF	DSG	EDV	BAL_EN	TC	TD	FC	FD			
IT Status (high)	0x0004	RSVD	RSVD	RSVD	OCVFR	LDM	RX	QMAX	VDQ			
Gauging Status	0x0030	NSFM	RSVD	SLPQMAX	QEN	VOK	RDIS	RSVD	REST			
Manufacturing ...	0x0030	CAL_EN	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD			
Manufacturing ...	0x0030	RSVD	PF_EN	LF_EN	FET_EN	GAUGE_EN	DSG_TEST	CHG_TEST	RSVD			
Safety Alert A+...	0x0000	RSVD	RSVD	OTD	OTC	RSVD	RSVD	RSVD	RSVD			
Safety Alert A+...	0x0000	RSVD	RSVD	RSVD	OC	RSVD	OCC	COV	CUV			

Log Panel

Transaction Log

Name	Cnd	Result	Read Addr
FET_EN	0x22	N/A	N/A
FET_EN	0x22	N/A	N/A
DISFET_L	0x20	N/A	N/A
GAUGE_EN	0x21	N/A	N/A
CHG_FET_L	0x1F	N/A	N/A



Battery Charger/Solar Panel

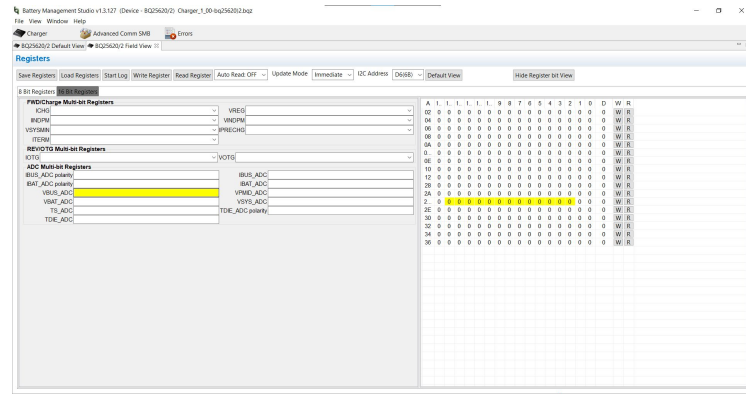
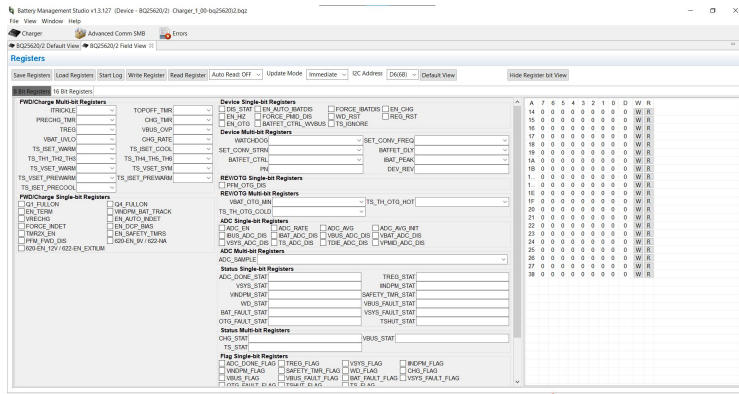
Cole Miller

Accomplishments since last presentation
27 hrs of effort

- Completed assembly of new board
- Validated board and met specs
- Begun integration test with Camden's board

Ongoing progress/problems and plans until the next presentation

- Assemble a list of registers for Jaein
- Remove old USB connector from solar panel
- Integrate to Patrick's and Camden's board



Battery Charger/Solar Panel Validation

Cole Miller

What Works

- Able to communicate with MCU
- Able to limit current and voltage
- Can work in boost mode
- ADC functionality works
- Able to integrate with Camden's board
- Meets 90% efficiency standard

Test Name	Input	Output
Standard Operation	5V 3A on VBUS	4.2 V 3.2A on VBAT
USB OTG Function	4.2V 3.2A on VBAT	5V 1.5A on VBUS
ILIM Register functionality	If input current > 1A	Output 1A
VLIM functionality	If Vin > 4.2V	Output 4.2V
PG Pin functionality	Good input	Low signal
STAT Pin Functionality	Charging	Low signal
STAT Pin functionality	Not Charging	High
System overvoltage Protection	Vin > Vsys_op	Vsys_fault_flag = 1
ILIM hardware functionality	Input current > 3A	Output current = 3A
Test name	Expectations	
I2C Commucation	Use SDA to communicate with EV2400	
Solar Panel VOC	Solar panel Voc < 18v	
Solar panel power	Can output 15W on a clear sunny day	
ILIM pin disable	Able to disable automatic limit current using register	
Switching speed	1.5 MHz on SWS pin	



Power MUX

Camden Pomeroy

Accomplishments since last presentation 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Assembled new PCB board • Debugged and blue wired a solution to subsystem • Fully validated new PCB board • Started integration testing two different PCB boards	• Further integration testing with physical PCB boards • Integrated PCB layout design • Research Solidworks software and design for battery pack casing

Power MUX Validation

Camden Pomeroy

What works:

- Input Voltage = Output Voltage
- UVLO/OVLO Operates as expected
- EN pin opens and shuts off properly
- TMUX fully validated
- Correct VBUS design
- Current draw does not overheat board

Number	VALIDATION PLAN	Input:	Excpeted Output:	Results	Status
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	4.5V of Vout when input was 5V	
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected	
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is Low	As expected	
4	PP1 PG (Bad signal)	if Vin < 2V	PG signal is High	As expected	
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is High	As expected	
6	PP2 PG (Bad signal)	if Vin < 2V	PG signal is Low	As expected	
7	PP1 UVLO Operational	if Vin < UVLO	No output	As expected	
8	PP2 UVLO Operational	if Vin < UVLO	No output	As expected	
9	PP1 OVLO Operational	if Vin > OVLO	No output	As expected	
10	PP2 OVLO Operational	if Vin > OVLO	No output	As expected	
11	SEL PP1	SEL is high	PP1 is active and PP2 is not	Sel pin would not poperly switch	
12	SEL PP2	SEL is low	PP2 is active and PP1 is not	Sel pin would not poperly switch	
13	ILM during startup	Inrush current of 4A	Limited current (<4 A) to ramp up	Could not get DC load to operate	
14	Seperate Power Paths	S1 active and both Vins 5V	Only PP1 Vout = 5V	As expected	
15	Seperate Power Paths	S2 active and both Vins 5V	Only PP2 Vout = 5V	As expected	
16	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected	



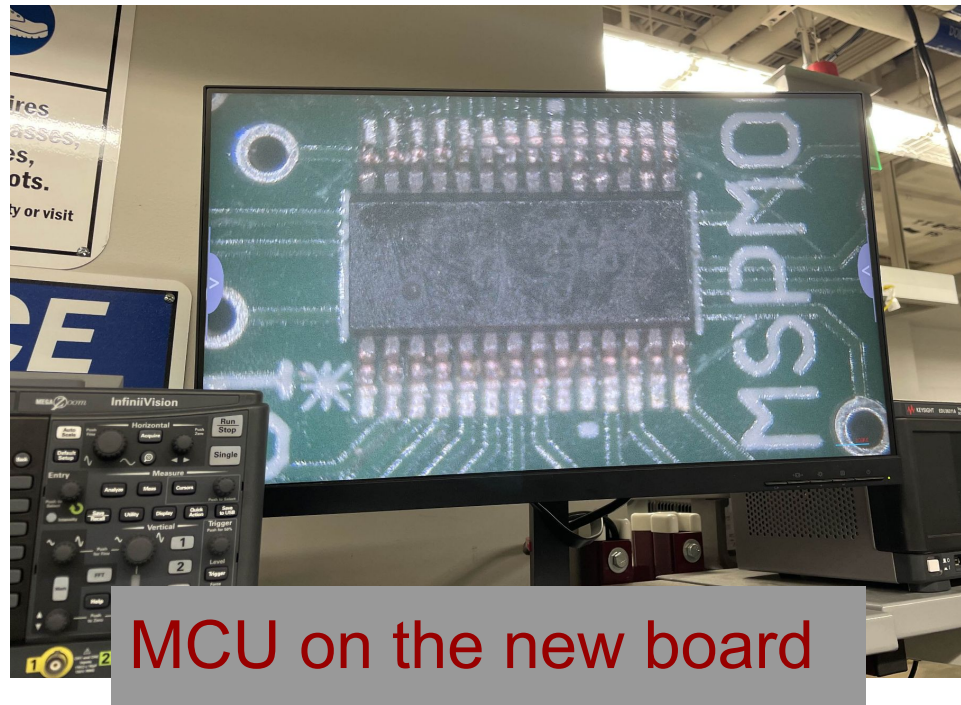
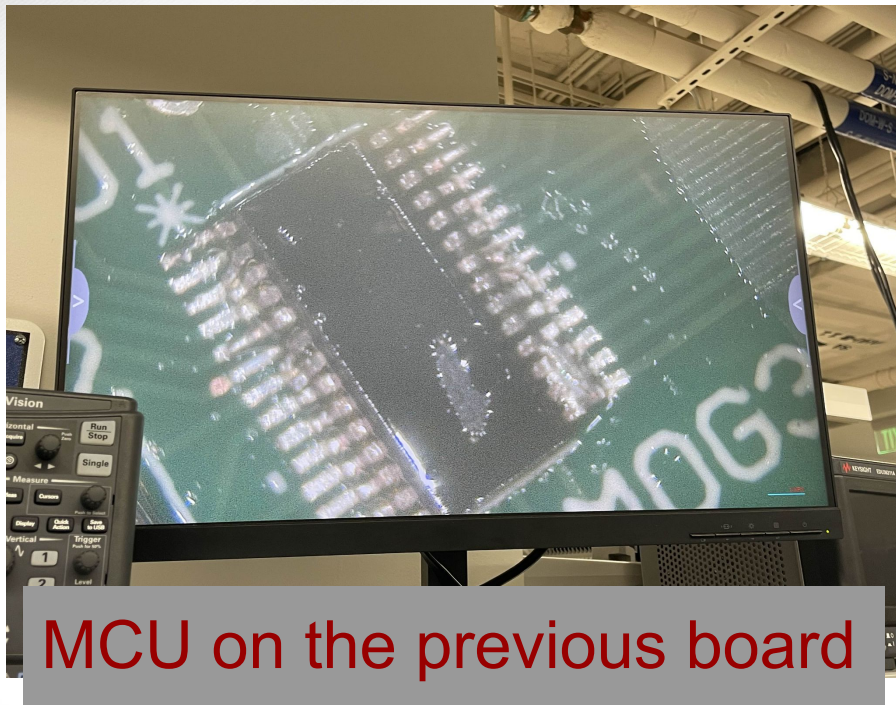
Microcontroller Unit

Jaein Cha

Accomplishments since last presentation 23 hrs of effort	Ongoing progress/problems and plans until the next presentation
– Ordered the PCB & Parts – Assembled the new board – PCB validation - tried to program the MCU on the custom board with a sample code	– Verify I2C functionality with other subsystem EVMs – Obtain specific information from other subsystems about required operations/data transfer/register addresses for firmware development – Make the MCU programmable

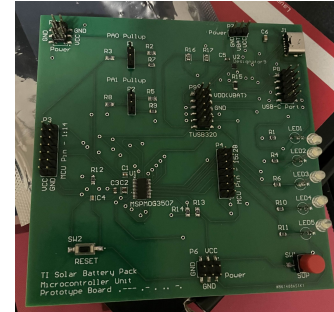
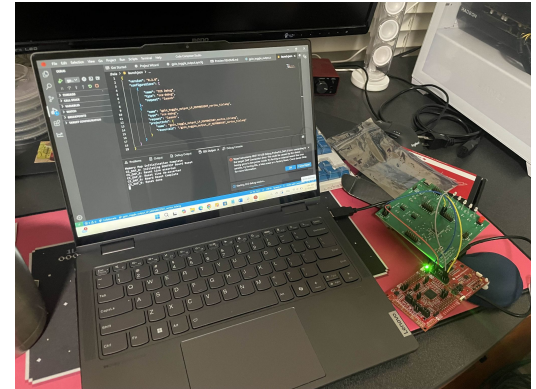
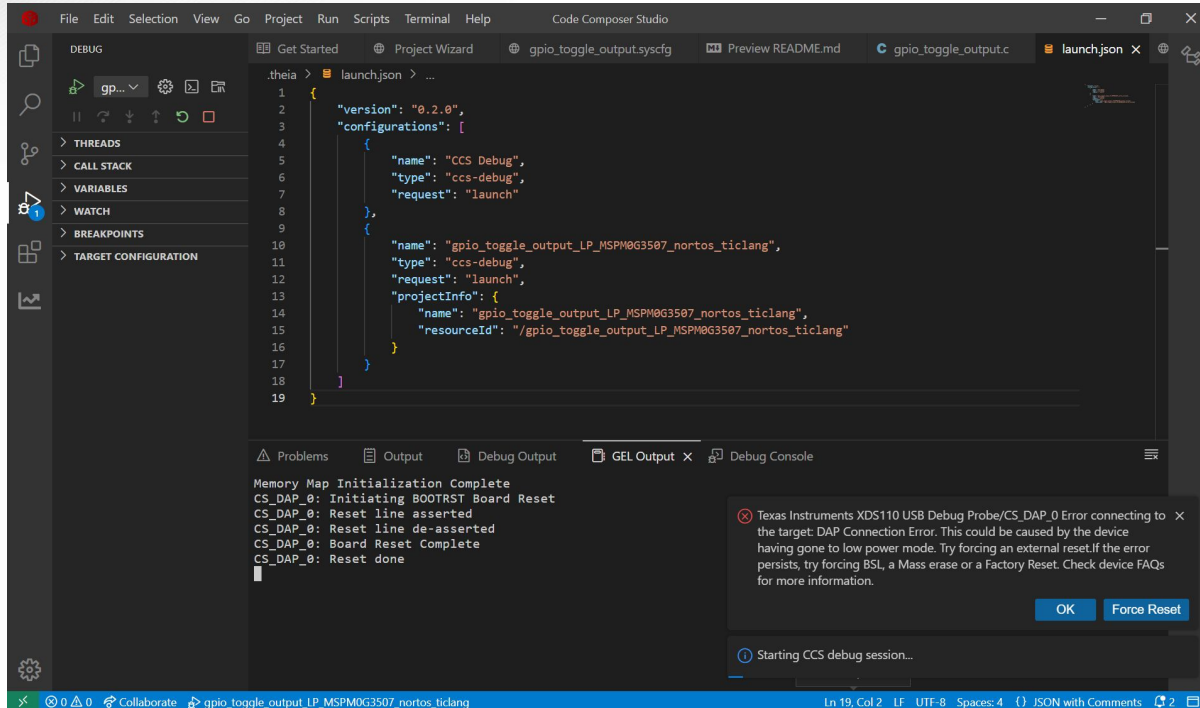
Microcontroller Unit

Jaein Cha



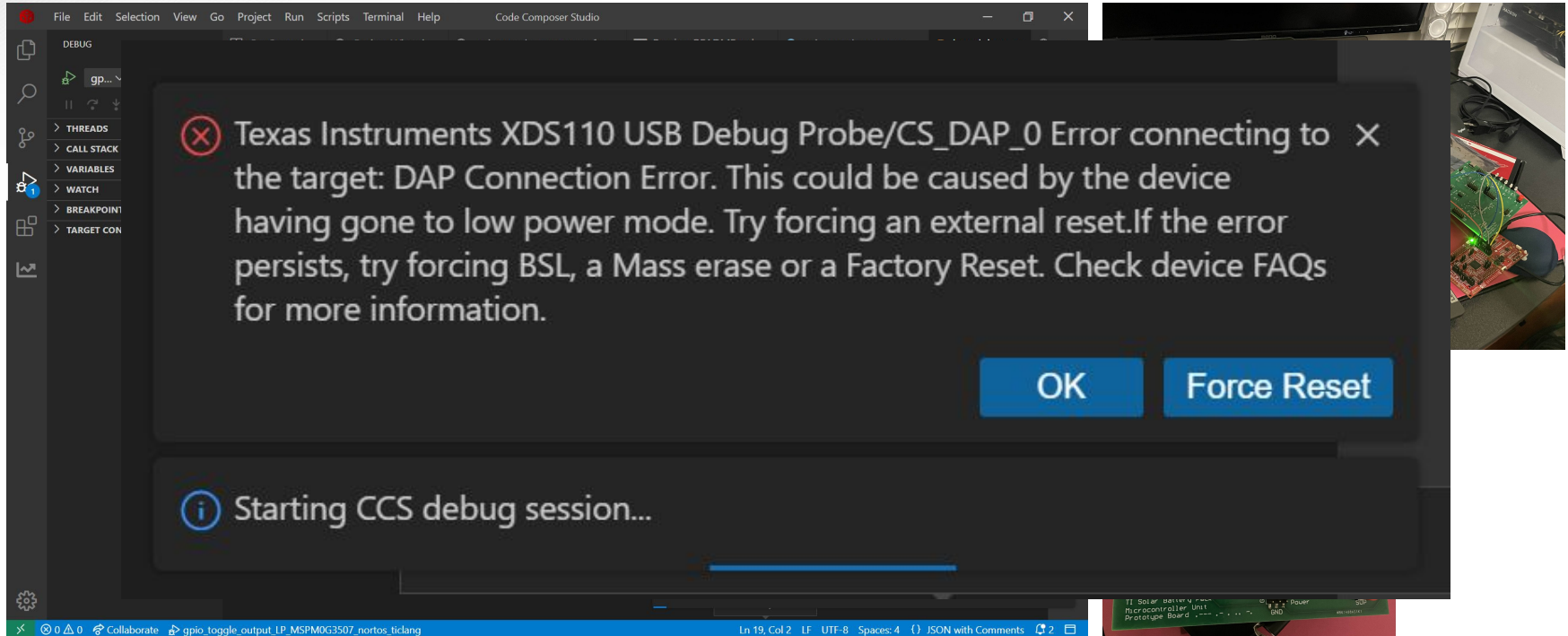
Microcontroller Unit

Jaein Cha



Microcontroller Unit

Jaein Cha



The screenshot shows the Code Composer Studio (CCS) interface. A large error dialog box is displayed in the center, with a red 'X' icon. The error message reads: "Texas Instruments XDS110 USB Debug Probe/CS_DAP_0 Error connecting to the target: DAP Connection Error. This could be caused by the device having gone to low power mode. Try forcing an external reset. If the error persists, try forcing BSL, a Mass erase or a Factory Reset. Check device FAQs for more information." Below the message are two buttons: "OK" and "Force Reset".

Below the error dialog, a status bar indicates "Starting CCS debug session...".

In the background, a microcontroller prototype board is visible, with a green LED lit, indicating it is powered on. The board is labeled "TI Stellaris V-Microcontroller Unit Prototype Board".

The bottom of the screen shows the CCS status bar with various icons and text: "Ln 19, Col 2 LF UTF-8 Spaces: 4 () JSON with Comments 2".



Parts Ordering Status

- All subsystem boards and parts are received

Execution & Plan

[illegible]



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Questions?